Research Documentation: Concrete Mix Scenarios

Submitted to: Nebraska Department of Transportation (NDOT)

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Overview:

The following four mix designs represent standard and high-performance concrete classes used by the Nebraska Department of Transportation (NDOT). These scenarios were selected to ensure the Python model can handle multiple different types of cement, water-cement ratios, as well as different aggregates

Concrete Mixes:

Scenario 1: Class 47B Paving (Standard)

- Cement Weight: 510 lb.
- Fly Ash Weight: 110 lb.
- Water/Cement Ratio: 0.44
- Target % Air Content: 6.5%
- Aggregate Split: 70% fine / 30% Coarse

Class 47B Paving is mainly used for highway pavement and general-purpose paving as laid out in the NDOT Standard Specifications Section 1002.02. This pavement's strength is its durability from freeze-thaw cycles due to the 6.5% air content.

Scenario 2: Class 47BD Bridge Deck (High Performance)

- Cement Weight: 590 lb.
- Fly Ash Weight: 130 lb.
- Silica Fume Weight: 40 lb.
- Water/Cement Ratio: 0.40
- Target % Air Content: 7.5%
- Aggregate Split: 40% Fine / 60% Coarse

Class 47 BD Bridge Deck concrete is used for Elevated bridge decks requiring low permeability and high durability (NDOT Standard Specifications Section 1002.02). The silica fume causes low permeability and the tighter water cement ratio will improve structural integrity.

Scenario 3: Class 47BR Revised/Optimized

- Cement Weight: 520 lb.
- Fly Ash Weight: 100 lb.

- Water/Cement Ratio: 0.40
- Target % Air Content: 6.0%
- Aggregate Split: 60% Fine / 40% Coarse (Well-graded/Optimized)

Class 47BR Concrete uses optimized pavement mixes which focus on dene aggregate packing to reduce shrinkage after the pour (NDOT Report on Optimized Aggregates Gradations).

Scenario 4: Class PR3 (High Early Strength)

- Cement Weight: 650 lb.
- Silica Fume Weight: 50 lb.
- Water/Cement Ratio: 0.38
- Target % Air Content: 6.5%
- Aggregate Split: 45% Fine / 55% Coarse

Class PR3 Pavement is used for repair where rapid strength gain is required for a road reopening (NDOT Standard Specifications Section 1002). The High Cementous weight causes a rapid hydration throughout the mix, and a lower water cement ratio can improve early strength.

Material Constants:

- SG Cement: 3.15
- SG Fly Ash: 2.30
- SG Silica Fume: 2.20
- SG Fine Aggregate: 2.62
- SG Coarse Aggregate: 2.66

References

Nebraska Department of Transportation (NDOT). (2017). *Standard specifications for highway construction*. Lincoln, NE: Nebraska Department of Transportation. <https://dot.nebraska.gov/media/g4qp4y0d/2017-specbook.pdf>.

Nebraska Department of Transportation (NDOT). (2023). *Concrete mix design submittal tool* [Excel workbook]. Lincoln, NE: Nebraska Department of Transportation. <https://dot.nebraska.gov/media/jp3paote/mix-design-submittal.xlsx>.

Nebraska Department of Transportation (NDOT). (2021). *Optimized aggregates gradations for portland cement concrete mix designs evaluation*. Final Report P336. Lincoln, NE: Nebraska Department of Transportation. <https://dot.nebraska.gov/media/3isfdv45/final-report-p336.pdf>